Classical Mechanics

Leonard Susskind

Core track, 2011 Fall Theoretical Minimum
Transcript-derived notes curated by LazyingArt LLC with Video2Book

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC with Video2Book.

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Chapter 1

What Counts as a Law of Motion?

Classical mechanics begins here with a question that is broader than any one orbit, any one falling stone, or any one charged particle. We want to know not only the specific laws governing particular systems, but also the general framework within which an acceptable law of motion must lie. Leonard Susskind therefore begins, quite deliberately, with absurdly simple worlds. The simplicity is not a detour. It is what allows the notions of state, law, determinism, conservation, and reversibility to emerge cleanly before the familiar machinery of mechanics is brought in. The present notes follow that unfolding closely, and the preserved board frames from LazyingArt LLC and Video2Book are used only where they steady a visible equation or diagram already motivated in the lecture.

1.1 Two Questions and a Minimal World

The lecture opens by separating two questions that are easy to mix together.

The first question is concrete: what are the laws for particular kinds of systems? A planet moving in the field of a heavy mass has one kind of law; a charged particle moving in a magnetic field has another. These are the specific laws of nature.

The second question is more structural: what counts as an allowable law at all? What must be true of a law of motion before we even begin to ask which particular system it describes? Susskind tells us immediately that this second question is the real target, while the first will merely supply illustrations.

So we begin with the smallest world that is still nontrivial. Let us take a coin lying on an abstract table. The only feature of the coin that matters is whether it is heads or tails. Thus the world contains one object with two configurations,

$$H \qquad \text{and} \qquad T.$$

An initial condition is just the choice of one of these configurations at the initial time. A law of motion is then a rule that tells us what happens next.

This order matters. First we decide what the possible configurations are. Only then do we ask how they evolve.

1.2 A Coin as a Dynamical System

For the coin, the configuration space is simply

$$\mathcal{C} = \{H, T\}.$$

Susskind first pauses over the vocabulary. This is our world. What is an initial condition? It is just the statement that the coin is heads or tails. What, then, is a law of motion? We are about to invent one.

He briefly remarks that there is an even simpler world: a one-sided coin. But that world is too simple, because nothing can happen in it. The two-state coin is the first place where the question becomes interesting.

Time is introduced in the simplest possible form. We imagine a stroboscopic world in which we look only at successive flashes, so the time variable is discrete:

$$t \in \mathbb{Z}.$$

The first law: nothing happens. The first law is as boring as possible:

$$H \to H, \qquad T \to T.$$

Heads remains heads, tails remains tails. This is a trivial law, but it is already a powerful one. Once the initial condition is known, the entire future is fixed. If the world begins in H, its history is

$$H, H, H, H, \ldots$$

and if it begins in T, its history is

$$T, T, T, T, \ldots$$

The second law: flip every step. There is only one other deterministic one-step law for a two-state system:

$$H \to T, \qquad T \to H.$$

At each flash of the stroboscope, the coin changes to the opposite state. Starting from heads, the history is

$$H, T, H, T, \ldots$$

and starting from tails, it is

$$T, H, T, H, \ldots$$

Only after the pictures are in place does Susskind introduce notation. Let

$$\sigma = \begin{cases} 1, & \text{for } H, \\ -1, & \text{for } T. \end{cases}$$

Then the two states are compressed into the two values of one variable. In that notation, the boring law becomes

$$\sigma(t+1) = \sigma(t), \tag{1.1}$$

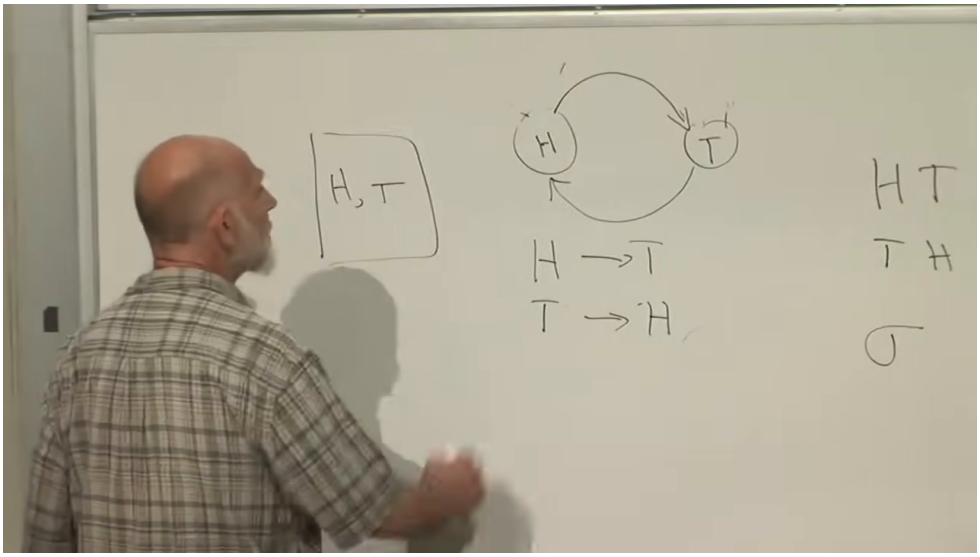


Figure 1.1: The board layout for the two-state coin system: the boxed state set at left, the directed two-state cycle at center, and the explicit update rules beneath.

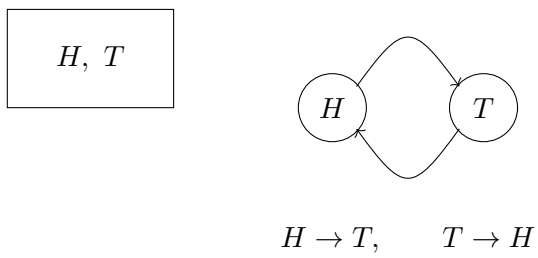


Figure 1.2: A clean reconstruction of the flip law.

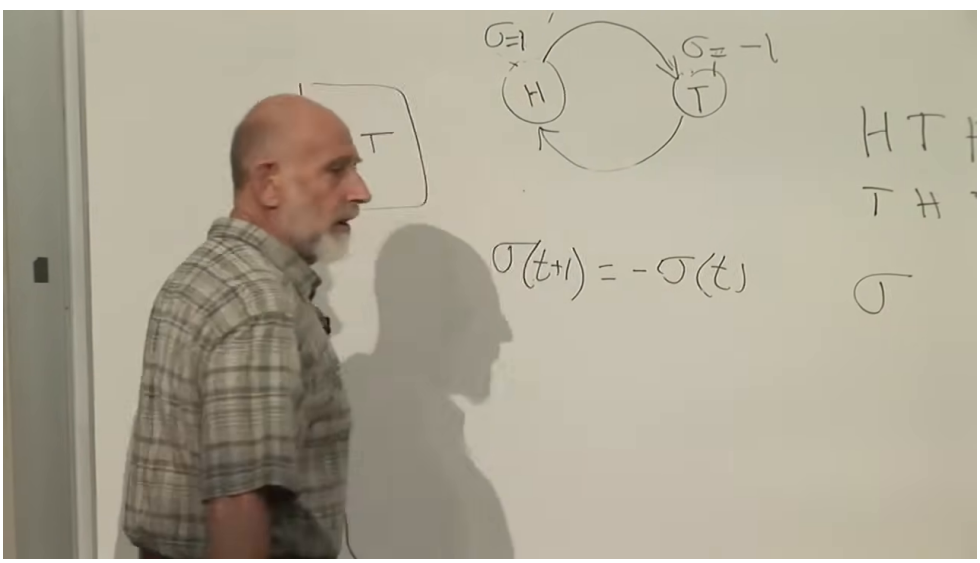


Figure 1.3: The same two-state system after the lecture upgrades it to sigma notation and writes the equation $\sigma(t+1) = -\sigma(t)$.

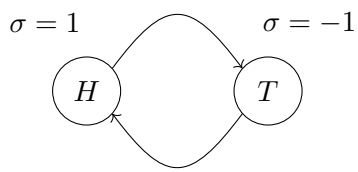


Figure 1.4: A clean reconstruction of the sigma-labeled state diagram.

and the flip law becomes

$$\sigma(t+1) = -\sigma(t). \tag{1.2}$$

This is the first important rhythm of the lecture: concept first, notation second. The symbol σ does not create the physics; it merely packages what was already clear from the update rule and the graph.

The coin itself is not the point. The point is that, for a closed system, a law of motion in classical mechanics is completely predictive. Once the law and the initial state are specified, the future is fixed.

1.3 From a Coin to a Die: Cycles, Relabeling, and Conservation

Now let us complicate the system, but not by much. Instead of a coin, take a die. The only fact about the die that matters for this discussion is that it has six possible states:

$$1, 2, 3, 4, 5, 6.$$

An initial condition is again just a choice of one of these states.

At this stage Susskind does not try to write equations. The graph is the right language. A law of motion is a directed graph on the state space, and the history of the system is found by iterating the arrows.

The most boring law is again the one in which nothing happens: every state points to itself. More interesting is a single six-cycle,

$$1 \to 2 \to 3 \to 4 \to 5 \to 6 \to 1. \tag{1.3}$$

One spoken line in the lecture is slightly garbled at this point, but the intended six-cycle is clear from the surrounding exposition and from the histories he derives from it.

He then writes down a more awkward-looking law,

$$1 \to 2, \qquad 2 \to 5, \qquad 5 \to 3, \qquad 3 \to 4, \qquad 4 \to 6, \qquad 6 \to 1,$$

but this is not really a new kind of law. It is still one cycle passing through every state. The difference is only a relabeling. If we were to redraw the nodes in a different order, the graph would look just like the first one.

So what genuinely distinguishes different laws? Not the names we assign the states, but the cycle structure.

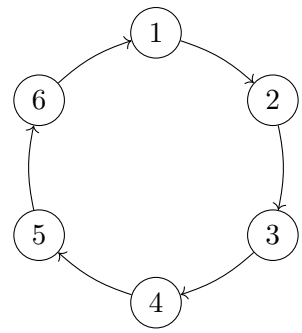


Figure 1.5: A law consisting of one six-cycle.

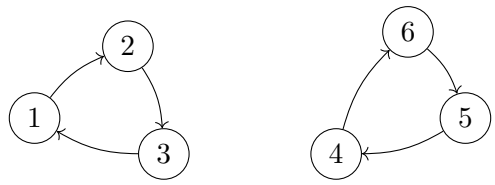


Figure 1.6: A law with two disconnected cycles.

A genuinely new possibility is a graph with more than one disconnected cycle. For example,

$$1 \to 2 \to 3 \to 1, \qquad 4 \to 6 \to 5 \to 4.$$

Now the die is still one system, but once the evolution begins, the state is trapped on one cycle or the other.

This is where conservation first appears. We may assign one label to the upper cycle and another to the lower. That label remains fixed as the state moves around inside its cycle.

Definition 1.1. A conserved quantity is a label on the state space that remains constant under the evolution.

In these finite examples, conserved quantities arise because the state space decomposes into disconnected pieces. Susskind then gives a further example with three cycles, including a fixed point, in order to emphasize that a conserved quantity need not specify the state completely. It may tell us only which orbit we belong to. The state changes; the conserved label does not.

1.3.1 Question & Answer

Question. What if the next step depends not only on the present configuration, but also on the previous one?

Answer. Then the present configuration alone was never the full state. The state must be enlarged until the law again becomes one-step deterministic.

This is a crucial point, and Susskind does not let it pass too quickly. Suppose, for the coin, that what happens next depends on both the present face and the immediately preceding face. Then the correct state space is no longer $\{H, T\}$. It is the four-element set

$$\{HH, HT, TH, TT\},$$

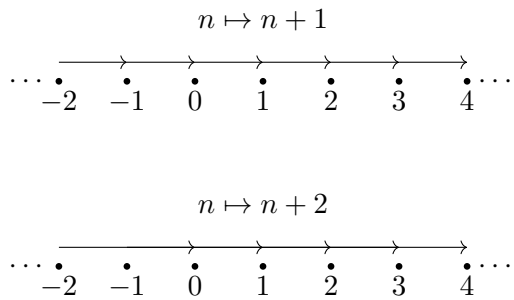


Figure 1.7: Two infinite-state laws on the integer lattice. The second preserves parity.

where, for example, HT means "heads now, tails one step ago."

So the right definition is not merely "what the system looks like now." It is sharper:

Definition 1.2. The state of a system consists of all the information that must be specified at one time in order to predict the future.

This principle will return in a much more serious form when we discuss particles. For the moment, the toy example has already done its work.

1.4 Infinite State Spaces and the Meaning of an Allowed Law

There is nothing in the discussion so far that requires the state space to be finite. We can just as well consider a particle that occupies one point on the infinite integer lattice,

$$n \in \mathbb{Z}.$$

The particle is still only one object. It simply has infinitely many possible states.

One possible law is

$$n(t+1) = n(t) + 1. \tag{1.4}$$

At each step the particle moves one unit to the right. If we think graphically, there is one endless orbit: going forward or backward, we pass through every integer.

A second law is

$$n(t+1) = n(t) + 2. \tag{1.5}$$

Now the odd integers communicate only with odd integers, and the even integers only with even integers. So parity is conserved.

A convenient encoding is

$$Q_{\text{parity}}(n) = \begin{cases} 0, & n \text{ even}, \\ 1, & n \text{ odd}. \end{cases}$$

Under $n \mapsto n+2$, this quantity is unchanged. Thus the language of cycles, conserved quantities, and deterministic evolution survives the move from finite to infinite state spaces.

But now the lecture reaches its sharpest conceptual pivot. The laws we have written so far have all shared an additional feature: not only do they predict the future, they also determine the past. Susskind now draws a law that is deterministic forward but not backward.

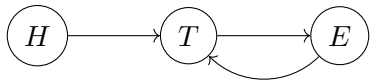


Figure 1.8: A future-deterministic law that is not reversible.

Take a three-state system with states H, T, and E (for "edge"). Let the arrows be

$$H \to T, \qquad T \to E, \qquad E \to T.$$

This law predicts the future perfectly. If we start at H, we get

$$H, T, E, T, E, \ldots$$

and if we start at T, we get

$$T, E, T, E, \ldots$$

But if all we know is that the present state is T, we do not know where we came from. The previous state could have been H or E.

1.4.1 Question & Answer

Question. Why is this law disallowed, if it still predicts the future?

Answer. Because classical mechanics demands more than forward predictability. It demands reversibility.

Susskind's test is beautifully simple: reverse every arrow. In the reversed graph, if we stand at T, we do not know which way to go next. There are two possibilities. So the reversed evolution is not predictive. The original law runs forward but cannot be run backward uniquely.

Definition 1.3. A law is reversible if the present state determines both the next state and the previous state. In the graph language, each state must have exactly one outgoing arrow and exactly one incoming arrow.

This is the real payoff of the first half of the lecture. A legal classical law is not merely deterministic into the future. It is deterministic into the past as well.

Susskind briefly remarks that probabilistic laws also fall outside the classical framework as he is defining it here. That larger comparison is postponed. For the present lecture, the essential distinction is already on the board: future determinism is not enough.

1.5 Exact Laws, Imperfect Initial Data, and the Pivot to Particles

At this point we might be tempted to say: classical mechanics predicts everything. Susskind immediately softens that slogan.

To predict the future we need two ingredients:

- the law of motion,

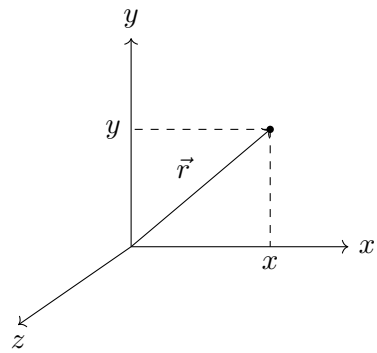


Figure 1.9: A point in a right-handed Cartesian coordinate system, together with its position vector.

- the initial condition.

A perfectly exact law is of no practical use if the initial condition is only imperfectly known.

In the toy systems this problem seems mild. One can imagine simply looking at the coin and reading its initial state exactly. But the real world is different. The degrees of freedom are continuous. Positions and velocities are real numbers, and no matter how many decimal places we measure, we never know them perfectly.

Thus there is a gap between exact determinism and practical predictability. Some systems magnify tiny initial errors. The weather is Susskind's example. If the atmospheric state is known only approximately, then after a long enough time the uncertainty blows up, and prediction fails in practice even though the underlying equations remain exact.

So the right classical statement is more careful. If we knew the initial conditions perfectly, then a true classical system would determine both future and past forever. If we know them only approximately, then we must ask how long that approximation remains useful.

This qualification does not undo determinism. It places it where it belongs: in the law itself, not in our practical ability to exploit it.

Only after making that distinction does Susskind turn to the more realistic world of particles. We are now ready to ask what the configurations of a point particle are. But before we do so, he stops and says exactly what the lecture needs next: a little mathematical preliminary.

1.6 Coordinates, Vectors, and the Algebra We Need

A coordinate system is simply a quantitative way to describe space. For the moment we take space to be three-dimensional, so a point is described by three coordinates,

$$(x, y, z), \qquad \text{or equivalently} \qquad (x_1, x_2, x_3).$$

We choose an origin, three mutually perpendicular axes, an orientation, and a system of units. Once the x- and y-axes are fixed, there remains one discrete choice for the direction of the z-axis, and the lecture adopts the standard right-handed convention.

Now let us recall what a vector is. A vector is an object with magnitude and direction. The first example is the position of a point relative to the origin, but Susskind is careful to say that the

notion of vector is more general than that. It may represent position, velocity, acceleration, an electric field, or any other quantity with length and direction.

Equally important is what a vector is not. It is not, in the first instance, tied to a particular place in space. If we slide the arrow without changing its length or direction, we have the same vector.

If a vector has components (v_x, v_y, v_z), then its magnitude is

$$|\vec{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2}. \tag{1.6}$$

The direction is encoded in the ratios of the components.

We can multiply a vector by an ordinary real number. If $c > 0$, the vector $c\vec{A}$ has the same direction as $\vec{A}$ and magnitude $c|\vec{A}|$. If $c < 0$, the direction is reversed. In particular, $-\vec{A}$ points opposite to $\vec{A}$.

We can also add vectors. The lecture presents three equivalent pictures:

- place the tail of $\vec{B}$ at the head of $\vec{A}$, and the third side of the triangle is $\vec{A} + \vec{B}$;
- place the tails of $\vec{A}$ and $\vec{B}$ together, complete the parallelogram, and take the diagonal;
- add components:
$$C_i = A_i + B_i.$$

This last form is often the most useful in calculation.

Susskind then asks the natural next question. If vectors can be added, subtracted, and multiplied by numbers, can they be multiplied with each other? The answer is yes, but not in the ordinary scalar sense. The simplest such product is the dot product.

Geometrically, $\vec{A} \cdot \vec{B}$ is obtained by taking the component of $\vec{A}$ along the axis defined by $\vec{B}$, and multiplying by the magnitude of $\vec{B}$. If θ is the angle between them, this gives

$$\vec{A} \cdot \vec{B} = |\vec{A}|\,|\vec{B}| \cos\theta. \tag{1.7}$$

Because this formula is symmetric in $\vec{A}$ and $\vec{B}$, it makes manifest that

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}.$$

In components, the same quantity is

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z. \tag{1.8}$$

In particular,

$$\vec{A} \cdot \vec{A} = |\vec{A}|^2. \tag{1.9}$$

This is precisely what makes the dot product useful. It lets us compute angles and test perpendicularity directly from components. If the dot product vanishes, the vectors are perpendicular. If it is negative, the angle between them is greater than 90°.

The law of cosines is then derived from the dot product. Here the live rhythm of the lecture matters: Susskind first says $\vec{C} = \vec{A} + \vec{B}$, then stops, corrects himself, and says that for the side drawn in the sketch we need

$$\vec{C} = \vec{A} - \vec{B}.$$

That correction should be preserved, because it shows the argument unfolding rather than arriving fully polished from nowhere.

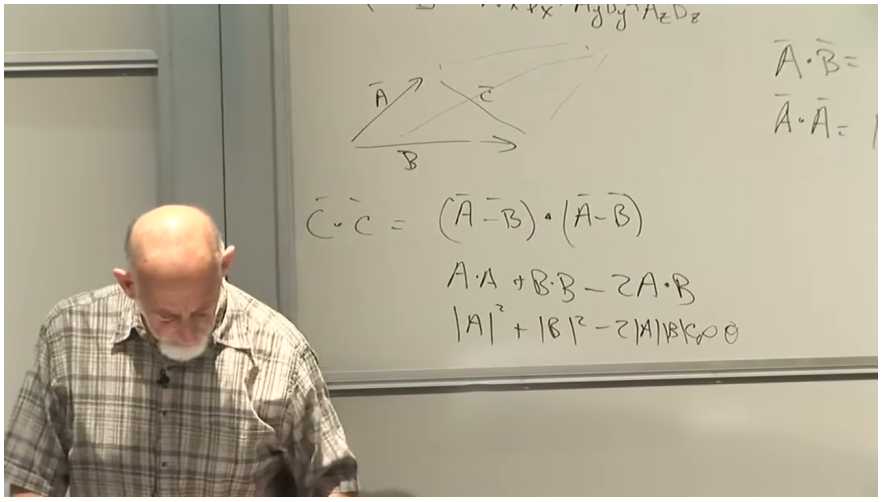


Figure 1.10: The lecture board at the vector-subtraction step leading to the law of cosines.

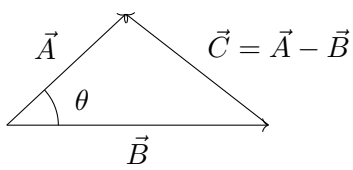


Figure 1.11: A clean reconstruction of the vector-subtraction triangle used in the derivation.

Now we compute:

$$
\begin{aligned}
|\vec{C}|^2 &= \vec{C} \cdot \vec{C} && (1.10)\\
&= (\vec{A} - \vec{B}) \cdot (\vec{A} - \vec{B}) && (1.11)\\
&= \vec{A} \cdot \vec{A} + \vec{B} \cdot \vec{B} - 2\vec{A} \cdot \vec{B} && (1.12)\\
&= |\vec{A}|^2 + |\vec{B}|^2 - 2|\vec{A}|\,|\vec{B}| \cos\theta. && (1.13)
\end{aligned}
$$

This is the law of cosines. The transcript momentarily garbles the last phrase as “cosine of the length,” but the mathematics is unambiguous: it is the cosine of the angle.

By this point the lecture has done its mathematical preparation. We have coordinates, vectors, component notation, vector addition, scalar multiplication, and the dot product. Now we can use them.

1.7 Position, Velocity, Acceleration, and Two Motion Examples

The position of a particle relative to the chosen origin defines a distinguished vector,

$$\vec{r}(t).$$

Its components are just the coordinates of the particle:

$$\vec{r}(t) = (x(t), y(t), z(t)).$$

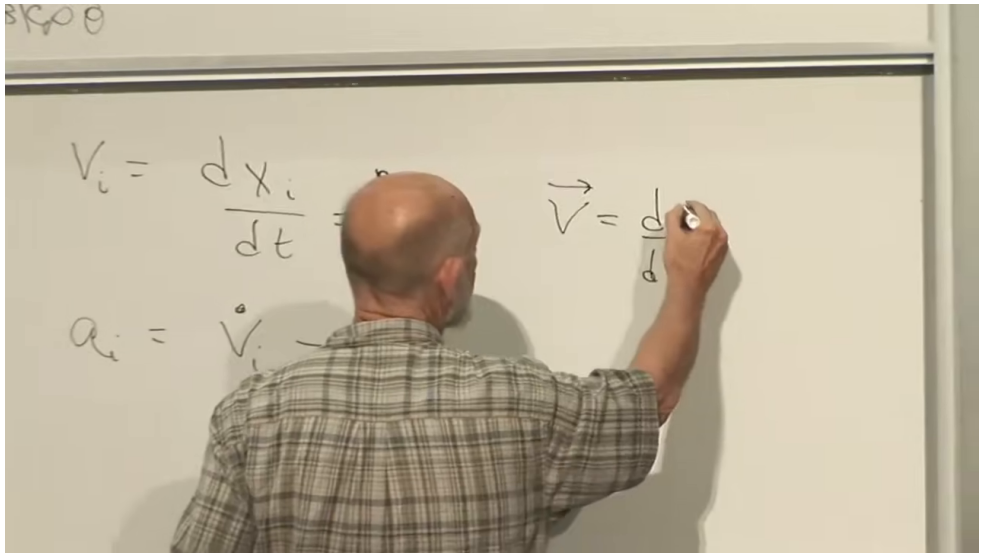


Figure 1.12: The lecture board moving from component notation to the compact vector statement $\vec{v} = d\vec{r}/dt$.

The notation r is traditional; Susskind remarks that it presumably comes from "radius," though here it simply means position vector.

Because classical mechanics concerns motion, $\vec{r}$ is in general a function of time. Thus the motion of a particle in three-dimensional space is encoded by the three functions

$$x(t), \qquad y(t), \qquad z(t).$$

The next concept is velocity. It is also a vector, and it need not point in the same direction as $\vec{r}$. Its definition is

$$v_i = \frac{dx_i}{dt} = \dot{x}_i, \tag{1.14}$$

or, in vector form,

$$\vec{v} = \frac{d\vec{r}}{dt} = \dot{\vec{r}}. \tag{1.15}$$

The dot notation is introduced at this point as a special shorthand for differentiation with respect to time only.

Acceleration is the rate of change of velocity. It too is a vector:

$$a_i = \frac{dv_i}{dt} = \dot{v}_i = \ddot{x}_i, \qquad \vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} = \ddot{\vec{r}}. \tag{1.16}$$

The screenshot shows uppercase V_i and X_i on the board. The surrounding lecture, however, does not insist on that convention, and for consistency we normalize to lower-case component notation in the text.

Now that the concepts are in hand, the lecture does exactly what it promised: let us work out an example or two.

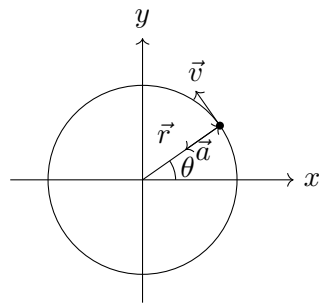


Figure 1.13: Uniform circular motion on the unit circle. The velocity is tangent to the orbit, while the acceleration points inward.

First example: motion along a line. If the particle moves only along the x-axis, then it is hardly worth speaking of vectors at all. We may simply write

$$x(t) = a + bt + ct^2. \tag{1.17}$$

At $t = 0$, the particle is at $x = a$. Differentiating once gives the velocity,

$$\dot{x}(t) = b + 2ct, \tag{1.18}$$

and differentiating again gives the acceleration,

$$\ddot{x}(t) = 2c. \tag{1.19}$$

So the acceleration is constant. This is uniformly accelerated motion, the prototype of motion in a constant gravitational field.

Second example: uniform circular motion. The final example is more revealing. Consider a particle moving on the unit circle in the xy-plane. We ignore the z-direction. Let the angle grow linearly with time:

$$\theta(t) = \omega t. \tag{1.20}$$

One full revolution corresponds to $\theta = 2\pi$, so the period is

$$T = \frac{2\pi}{\omega}. \tag{1.21}$$

The transcript momentarily stumbles at the period formula, but the final expression is clear and standard.

On the unit circle, the position components are

$$x(t) = \cos(\omega t), \qquad y(t) = \sin(\omega t). \tag{1.22}$$

Differentiating once, we find the velocity components:

$$v_x(t) = -\omega \sin(\omega t), \qquad v_y(t) = \omega \cos(\omega t). \tag{1.23}$$

1.7.1 Question & Answer

Question. It looks obvious that the velocity is perpendicular to the position vector. How do we prove it?

Answer. We prove it with the dot product:

$$\begin{aligned}\vec{r}\cdot\vec{v} &= xv_x + yv_y && (1.24)\\ &= \cos(\omega t)\big(-\omega\sin(\omega t)\big) + \sin(\omega t)\big(\omega\cos(\omega t)\big) && (1.25)\\ &= 0. && (1.26)\end{aligned}$$

Since the dot product vanishes, $\vec{r}$ and $\vec{v}$ are perpendicular.

This is one of the lecture's nicest little moments. First we trust the geometry. Then we check it algebraically. The calculation does not replace the picture; it confirms it.

Differentiating once more, we obtain the acceleration:

$$a_x(t) = -\omega^2\cos(\omega t), \qquad a_y(t) = -\omega^2\sin(\omega t). \tag{1.27}$$

Therefore

$$\vec{a} = -\omega^2\vec{r}. \tag{1.28}$$

So the acceleration is parallel to $\vec{r}$ but points in the opposite direction. It is inward, toward the origin. The spoken phrase "centrifugal acceleration" in the lecture is evidently a slip; the mathematics is perfectly clear.

Finally, the magnitudes are

$$\begin{aligned}|\vec{v}|^2 &= v_x^2 + v_y^2 && (1.29)\\ &= \omega^2\sin^2(\omega t) + \omega^2\cos^2(\omega t) && (1.30)\\ &= \omega^2, && (1.31)\end{aligned}$$

hence

$$|\vec{v}| = \omega.$$

Likewise,

$$\begin{aligned}|\vec{a}|^2 &= a_x^2 + a_y^2 && (1.32)\\ &= \omega^4\cos^2(\omega t) + \omega^4\sin^2(\omega t) && (1.33)\\ &= \omega^4, && (1.34)\end{aligned}$$

so

$$|\vec{a}| = \omega^2.$$

The closing example is not an afterthought. It is where the formalism becomes mechanics again: position, velocity, acceleration, orthogonality, inward acceleration, and the concrete role of ω.

1.8 Summary

The lecture begins by asking what an allowable law of motion must look like, and it answers that question before ever discussing a planet or a force. Coins, dice, and integer lattices are enough to reveal the structure: a state space, an initial condition, an update rule, deterministic evolution, conserved quantities, and, crucially, reversibility.

Only after that conceptual warm-up does the lecture move to particles in space. There we build the coordinate and vector language that mechanics will need: components, magnitudes, vector addition, the dot product, and the law of cosines. With those tools in hand, position, velocity, and acceleration follow naturally, and the two final examples show the formalism at work. The next step, exactly as Susskind says at the end of the lecture, is to return to particles and ask the full dynamical question: what is the space of initial conditions, and what laws move a particle from one instant to the next?

Chapter 2

Laws of Motion, Phase Space, and Conservation

We begin, as Susskind does, with a wrong law. That is not antiquarian stage-setting. It is a way of asking what a law of motion is supposed to accomplish. A law should tell us how to move from one instant to the next, and a fundamental law ought also to preserve enough information that the past is not washed away by the future. Aristotle's law is close enough to ordinary friction-dominated experience to feel plausible, and precisely for that reason it is a good place to begin. By watching it fail, we see more sharply what Newton changes.

2.1 Aristotle's Useful Wrong Law

Aristotle lived in a world where friction is everywhere. Push an object and it moves; stop pushing and it stops. The heavier the object, the harder it is to give it a certain speed; the harder one pushes, the faster it goes. Out of that everyday experience comes the deliberately wrong law

$$\vec{F} = m\vec{v}. \tag{2.1}$$

Susskind writes it down with some emphasis: this is the wrong equation. We are not to memorize it. We are to study it.

The point is not merely that Aristotle missed ice skating. The point is that he took force to be the source of motion itself, whereas Newton will make force the source of change of motion. But before correcting Aristotle, the lecture asks a more interesting question: in what sense is this even a law?

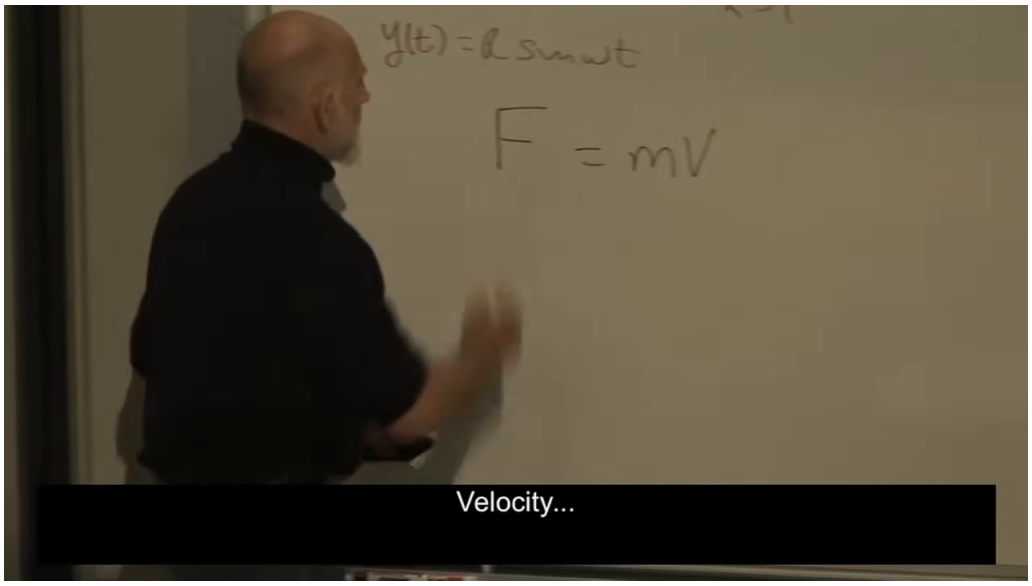
2.2 From Force to an Update Rule

To make that question precise, we pass at once to one dimension:

$$F = m\dot{x}. \tag{2.2}$$

Now we enter what Susskind calls a stroboscopic world. Time is broken into small intervals of size δ, and the velocity is approximated by

$$\dot{x}(t) \approx \frac{x(t+\delta) - x(t)}{\delta}. \tag{2.3}$$



Cleaning up the live board algebra, Aristotle's law becomes

$$F(t) = m\,\frac{x(t+\delta) - x(t)}{\delta}, \tag{2.4}$$

$$x(t+\delta) = x(t) + \frac{\delta}{m}F(t). \tag{2.5}$$

That is already a law of motion in a perfectly recognizable sense. If we know where the particle is at time t, and if we know the force, then the law tells us where it will be at time $t+\delta$.

This is the first important pivot of the lecture. A law becomes intelligible only when it becomes an update rule.

We can also read off the no-force case immediately. If $F = 0$, then $\dot{x} = 0$, so the object simply sits there:

$$F = 0 \qquad \Rightarrow \qquad \dot{x} = 0. \tag{2.6}$$

In Aristotle's world, no force means no motion. The future is static, and so is the past.

2.3 Aristotle's Spring and the Suspicion of Irreversibility

Susskind now chooses the same example that later will be used again under Newton: a spring force. Suppose

$$F(x) = -kx. \tag{2.7}$$

Insert this into Aristotle's update rule:

$$x(t+\delta) = x(t) - \frac{k\delta}{m}x(t) = \left(1 - \frac{k\delta}{m}\right)x(t). \tag{2.8}$$

If δ is small, then $1 - k\delta/m$ is slightly less than 1. So each time step multiplies the position by the same common factor. The motion is not oscillatory at all. The particle moves toward the origin, then closer, then closer still, shrinking by the same factor each step.

The continuous version says the same thing more cleanly. The sign in the spoken transcript wobbles a bit, but the intended equation is the one consistent with $F = -kx$:

$$m\dot{x} = -kx. \tag{2.9}$$

Its solution is

$$x(t) = x(0)e^{-kt/m}. \tag{2.10}$$

So under Aristotle's law a displaced spring simply drifts back to equilibrium, slower and slower, exponentially decaying toward the origin. It does not swing through. It does not remember where it started except through an exponentially fading trace.

2.3.1 Question & Answer

In what sense does Aristotle's law fail to predict the past?

Not in the trivial sense that the differential equation cannot be manipulated backward on paper. With infinite precision one can formally reconstruct earlier values. The lecture's point is subtler and more physical. Distinct initial conditions all flow into the same final resting state. If at a later time the particle is sitting essentially at the origin, then within any finite observational accuracy we can no longer tell where it came from. Initial distinctions are not preserved. Different histories collapse onto the same practical future.

This is why Susskind says Aristotle's law goes in the wrong direction from the point of view of reversibility. It predicts the future well enough, but it steadily loses information about the past.

2.4 Newton's Law, Inertial Frames, and What the Law Really Says

At this point the lecture turns. We now know what was really wrong with Aristotle. Force is not what makes an object move; force is what changes its state of motion. Friction itself is a force. In ordinary life the force from our hand often balances friction, and that balance can masquerade as a force-velocity law. But that is not a fundamental law.

Newton's second law is

$$F = m\ddot{x}. \tag{2.11}$$

Before using it as a dynamical rule, Susskind pauses over inertial frames. The pause is brief but important. One should not be satisfied with the circular slogan that Newton's laws are true in inertial frames and inertial frames are the frames in which Newton's laws are true. The real statement is that there exist frames of reference in which Newton's laws hold. Those are the inertial frames. Other frames, accelerating or rotating relative to them, require extra structure.

He also pauses over the operational meaning of force and mass. A kilogram mass is identified by comparison with a standard mass. A spring balance gives meaning to a unit of force by a fixed amount of stretch. If one identical spring produces one unit of force, then two identical springs in parallel produce two units, and the acceleration doubles. If we keep the force fixed and double the mass, the acceleration is halved. In this way the statement

$$a \propto F, \qquad a \propto \frac{1}{m} \tag{2.12}$$

is not empty symbolism. It is the content of experiment.

And now the lecture asks the same question it asked of Aristotle's law: is Newton's equation predictive?

2.5 From Second Order to Phase Space

If we discretize the second derivative in the standard way, we obtain

$$\ddot{x}(t) \approx \frac{x(t+\delta) - 2x(t) + x(t-\delta)}{\delta^2}. \tag{2.13}$$

The board algebra at this point is corrected in real time and is not worth following literally line by line. What matters is the cleaned update formula:

$$x(t+\delta) = 2x(t) - x(t-\delta) + \frac{\delta^2}{m}F(t). \tag{2.14}$$

Now a problem appears. To determine the next position, it is not enough to know the present position. We also need to know the earlier one. So Newton's law seems, at first glance, less predictive than Aristotle's.

2.5.1 Question & Answer

If Newton's law is second order, why does it still count as a predictive law of motion?

Because the apparent defect lies not in Newton's law but in the way we have described the state. "Knowing everything" about a system cannot mean merely knowing its position. It must mean knowing whatever is needed to predict the future. Newton's law teaches us that the right initial data are position together with velocity. The second-order law is not a failure of predictiveness; it is a clue that the space of states is larger than configuration space alone.

That larger state space emerges when we introduce momentum:

$$p = m\dot{x}. \tag{2.15}$$

Susskind writes this on the board and immediately rewrites Newton's law in terms of it.

Since the mass is taken to be constant in Newtonian mechanics, we have

$$\dot{p} = F, \tag{2.16}$$

or in vector form,

$$\vec{F} = \frac{d\vec{p}}{dt}. \tag{2.17}$$

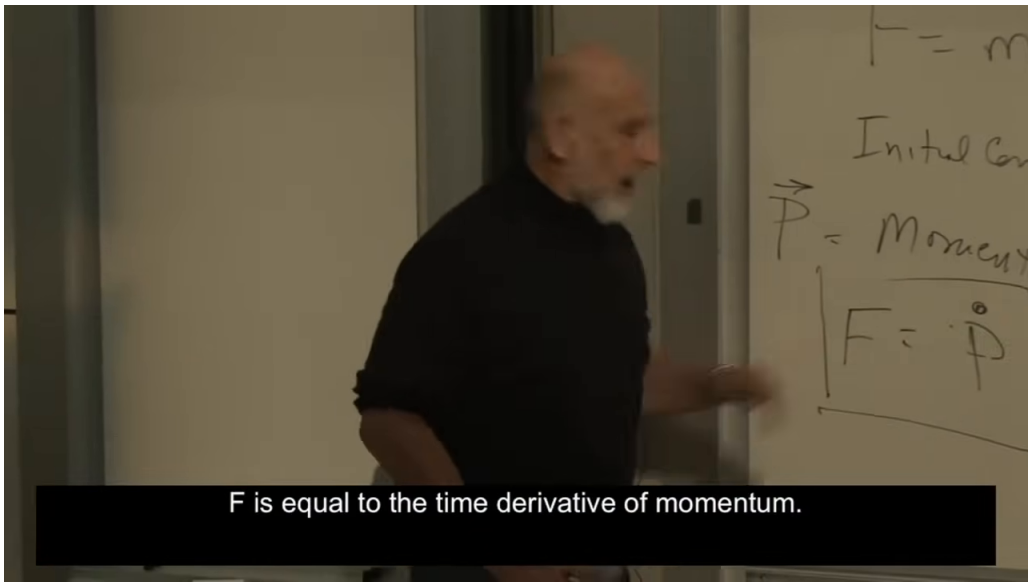
The second law has become one first-order equation. The definition of momentum gives the other:

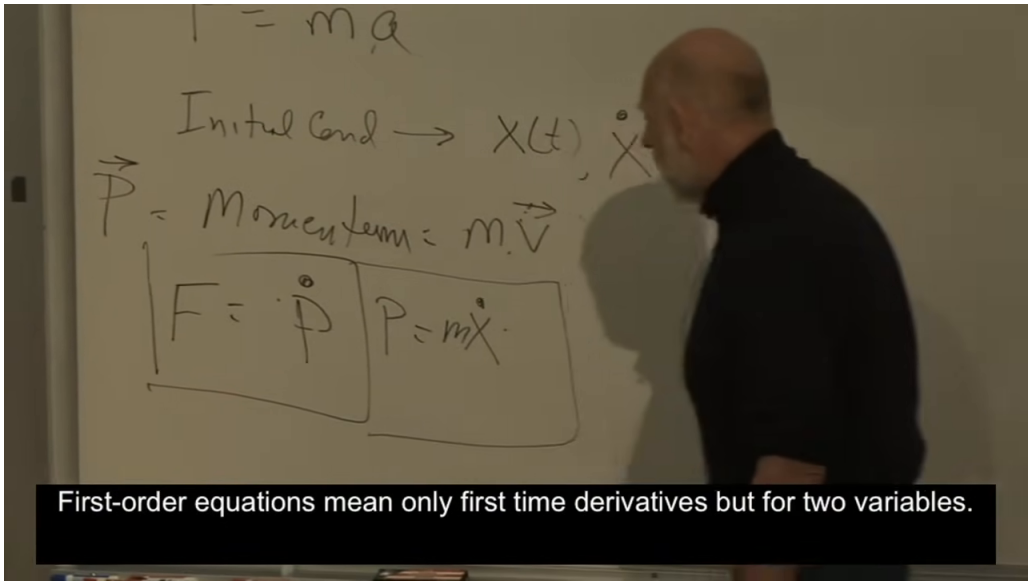
$$\dot{x} = \frac{p}{m}. \tag{2.18}$$

So Newtonian mechanics in one dimension may be written as the pair

$$\dot{p} = F, \tag{2.19}$$

$$\dot{x} = \frac{p}{m}. \tag{2.20}$$


F is equal to the time derivative of momentum.


Initial Cond → X(t)
P = Momentum = m.V
F = ṗ
P = mẋ
First-order equations mean only first time derivatives but for two variables.

In discrete language these become update rules,

$$p(t+\delta) = p(t) + \delta\, F(t), \tag{2.21}$$

$$x(t+\delta) = x(t) + \frac{\delta}{m} p(t), \tag{2.22}$$

to first order in δ. One equation tells us how the momentum changes if we know the force. The other tells us how the position changes if we know the momentum. If the force is given as a function of position rather than explicitly as a function of time, nothing essential changes: once x and p are known at an instant, the force is known too, and the update goes through.

The state of the particle is therefore not just x, but

$$(x, p), \tag{2.23}$$

or equivalently $(x, \dot{x})$. This enlarged space of initial data is phase space. The lecture stresses that point very strongly: phase space is the space of what one must know in order to predict the future.

2.6 The Harmonic Oscillator and Time Reversal

Now Susskind goes back to the spring and does the analogous Newtonian problem. This is not just another example. It is the same example under a different law, and therefore it reveals the structural difference between Aristotle and Newton with great clarity.

For the spring we have

$$m\ddot{x} = -kx. \tag{2.24}$$

To keep the logic clear, let us set $m = 1$ and $k = 1$. Then

$$\ddot{x} = -x. \tag{2.25}$$

The solutions are sines and cosines. A convenient form is

$$x(t) = c\cos(t - t_0), \tag{2.26}$$

where c and t_0 are the two constants of integration. Differentiating gives

$$\dot{x}(t) = -c\sin(t - t_0). \tag{2.27}$$

Since $m = 1$, the momentum is equal to the velocity, so

$$p(t) = -c\sin(t - t_0). \tag{2.28}$$

The sign of the trigonometric function is not the main point; the invariant relation is. Compute

$$x^2 + p^2 = c^2\cos^2(t - t_0) + c^2\sin^2(t - t_0) = c^2. \tag{2.29}$$

Thus the motion in phase space lies on a circle:

$$x^2 + p^2 = c^2. \tag{2.30}$$

That is the decisive contrast with Aristotle's spring. Under Aristotle's law all trajectories drift into the origin. Under Newton's law different initial conditions live on different cycles. Large

displacements give large circles, small displacements give small circles, but no trajectory collapses onto any other. The past is not lost.

Project the motion onto the x-axis and we see the ordinary oscillation back and forth. Project it onto the p-axis and we see the momentum oscillate as well. When x is maximal, $p = 0$. When the particle passes through the origin, the momentum is maximal. Position and momentum are exchanged back and forth, but the orbit in phase space remains fixed. In this normalization all circles are traversed with the same period; only the radius and the starting point change.

The lecture then gives another view of reversibility. Under time reversal,

$$t \mapsto -t, \qquad \dot{x} \mapsto -\dot{x}, \qquad \ddot{x} \mapsto \ddot{x}. \tag{2.31}$$

A first derivative changes sign, but a second derivative does not. So Aristotle's law is not invariant under reversal of time, whereas Newton's law is. If we film a Newtonian motion and run the film backward, we obtain another motion allowed by the same law.

A student asks why reversibility should be expected at all. Susskind's answer is deliberately modest. We are looking at successful laws of physics, searching for common structural features, and reversibility appears as one of them. At a deeper level the logical order runs the other way: Newtonian mechanics is an approximation emerging from more fundamental physics. But in this course we are proceeding from simple principles upward, and reversibility is one of the principles worth tracking.

2.7 Conservation Laws: Momentum, Energy, and Potential

The lecture now broadens from reversibility to conserved quantities. The harmonic oscillator has already suggested the idea: motion stays on one cycle and never jumps to another. That means there must be a label attached to the orbit. Susskind next derives two such labels, first momentum and then energy.

We begin with Newton's third law. Let $\vec{F}_{ij}$ denote the force on particle i due to particle j. Then

$$\vec{F}_{ij} = -\vec{F}_{ji}. \tag{2.32}$$

Susskind also remarks that real interparticle forces are directed along the line joining the particles, but he explicitly postpones that part of the discussion to the later treatment of angular momentum. For momentum conservation, the crucial point is equal and opposite pairs.

For a closed system of particles,

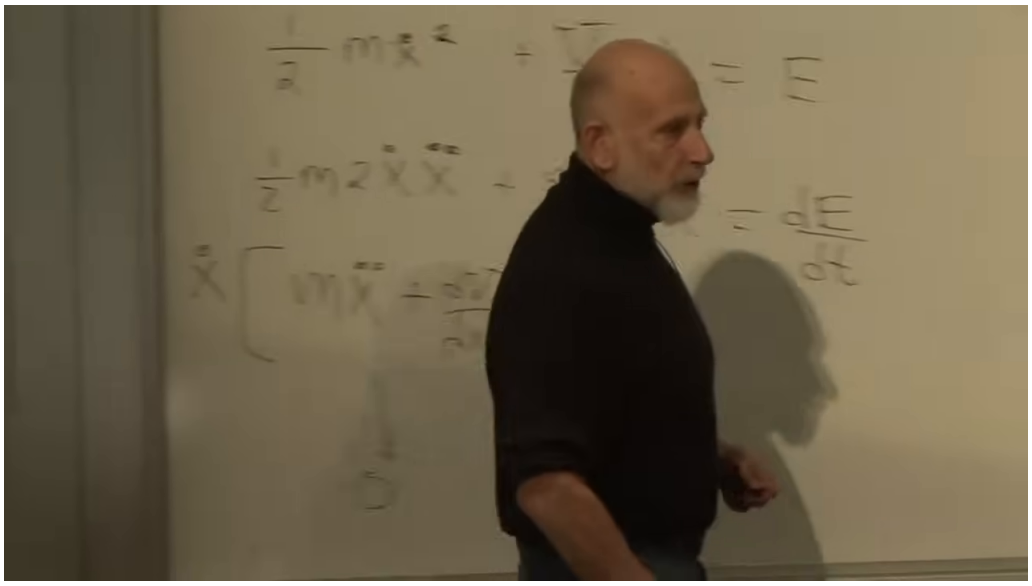
$$\dot{\vec{p}}_i = \sum_{j \neq i} \vec{F}_{ij}. \tag{2.33}$$

Now sum over all i:

$$\frac{d}{dt} \sum_i \vec{p}_i = \sum_i \sum_{j \neq i} \vec{F}_{ij}. \tag{2.34}$$

Each pair $\vec{F}_{ij}$ and $\vec{F}_{ji}$ appears once each and cancels. Therefore

$$\frac{d\vec{P}_{\text{tot}}}{dt} = 0, \tag{2.35}$$



where

$$\vec{P}_{\text{tot}} = \sum_i \vec{p}_i. \tag{2.36}$$

So total momentum is conserved.

After that first conservation law, the lecture returns to a single particle and asks for energy conservation. We begin in one dimension, with a force that depends only on position:

$$F = F(x). \tag{2.37}$$

Susskind is careful here. "Time independent" means not that the force cannot vary in time, but that its time dependence comes only through the particle's changing position. If the particle is at position x, the force is $F(x)$, no matter when that occurs.

In one dimension, for a sufficiently well-behaved force, we can always write

$$F(x) = -\frac{dV(x)}{dx}, \tag{2.38}$$

with

$$V(x) = -\int^x F(u)\,du + \text{const}. \tag{2.39}$$

The minus sign is a convention, but it is a useful one: it makes the force point toward decreasing potential. The lecture asks us to picture a particle moving along the x-axis on a terrain of hills and valleys whose height is $V(x)$. The steeper the hill, the larger the force; the force points downhill.

Now define the energy

$$E = \frac{1}{2}m\dot{x}^2 + V(x). \tag{2.40}$$

Here the first term is kinetic energy and the second is potential energy. Susskind emphasizes that, for present purposes, "energy" is simply the name we give to a conserved quantity if it turns out to be conserved.

Let us differentiate E:

$$\frac{dE}{dt} = \frac{d}{dt}\left(\frac{1}{2}m\dot{x}^2\right) + \frac{dV}{dt} \tag{2.41}$$

$$= m\dot{x}\,\ddot{x} + \frac{dV}{dx}\dot{x} \tag{2.42}$$

$$= \dot{x}\left(m\ddot{x} + \frac{dV}{dx}\right). \tag{2.43}$$

Now Newton's law says $m\ddot{x} = F$, and the potential relation says $F = -dV/dx$. Therefore

$$m\ddot{x} + \frac{dV}{dx} = 0, \tag{2.44}$$

so

$$\frac{dE}{dt} = 0. \tag{2.45}$$

That is energy conservation in one dimension. As the particle rolls downhill, V decreases while the kinetic term increases; as it climbs, the opposite occurs. The proof is simply the exact version of that intuitive trade-off.

For the spring force,

$$F = -kx, \tag{2.46}$$

we obtain

$$\frac{dV}{dx} = kx, \tag{2.47}$$

so

$$V(x) = \frac{k}{2}x^2. \tag{2.48}$$

The total energy of the oscillator is therefore

$$E = \frac{1}{2}m\dot{x}^2 + \frac{k}{2}x^2. \tag{2.49}$$

2.7.1 Question & Answer

Is every force field the gradient of a potential energy?

In one dimension, for nice enough $F(x)$, essentially yes: any such function can be integrated, and so one can define a potential. But in several dimensions the situation changes completely. If we are given three independent force components F_1, F_2, F_3, there is no mathematical necessity that they arise as partial derivatives of a single scalar V. In one dimension the existence of V is nearly automatic; in several dimensions it is a genuine physical assumption.

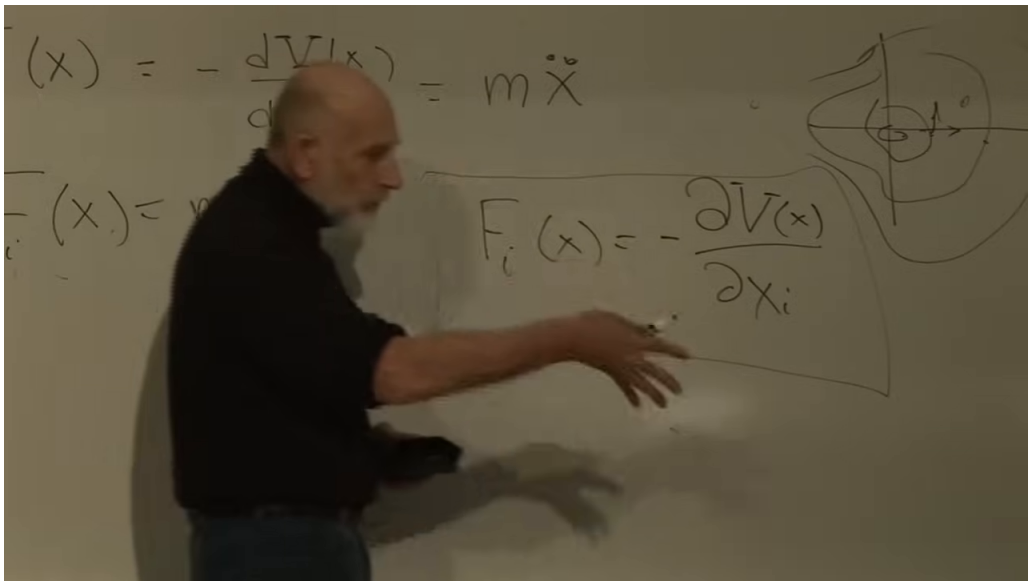
That is exactly the obstacle Susskind wants to raise before resolving it by postulate.

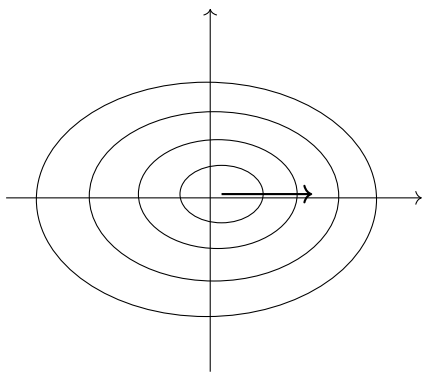
For one particle in several coordinates $\mathbf{x} = (x_1, \ldots, x_n)$, Newton's law reads componentwise

$$F_i(\mathbf{x}) = m\ddot{x}_i. \tag{2.50}$$

What we would like to assume is

$$F_i(\mathbf{x}) = -\frac{\partial V(\mathbf{x})}{\partial x_i}, \tag{2.51}$$





or, more compactly,

$$\vec{F} = -\nabla V. \tag{2.52}$$

This says that the force points in the direction of steepest descent of the potential. The contour-map picture on the board is precisely the picture of a scalar potential together with the downhill direction.

Once we assume this conservative-force form, the multidimensional energy is

$$E = \frac{1}{2} m \sum_i \dot{x}_i^2 + V(x_1, \ldots, x_n). \tag{2.53}$$

Its time derivative is

$$\frac{dE}{dt} = m \sum_i \dot{x}_i \ddot{x}_i + \frac{dV}{dt} \tag{2.54}$$

$$= m \sum_i \dot{x}_i \ddot{x}_i + \sum_i \frac{\partial V}{\partial x_i} \dot{x}_i \tag{2.55}$$

$$= \sum_i \dot{x}_i \left(m\ddot{x}_i + \frac{\partial V}{\partial x_i} \right). \tag{2.56}$$

If $m\ddot{x}_i = F_i$ and $F_i = -\partial V/\partial x_i$, then each bracket vanishes:

$$\frac{dE}{dt} = 0. \tag{2.57}$$

So energy is conserved in several dimensions as well, but now we must admit that we have paid a price. Unlike the one-dimensional case, we have made a real physical assumption: that the force components come from one scalar potential.

The many-particle generalization is then only a matter of bookkeeping. We flatten all coordinates of all particles into a single list $x_1, \ldots, x_{3N}$, allow corresponding masses m_i, and define

$$E = \sum_{i=1}^{3N} \frac{1}{2} m_i \dot{x}_i^2 + V(x_1, \ldots, x_{3N}). \tag{2.58}$$

If the force components satisfy

$$F_i = -\frac{\partial V}{\partial x_i}, \tag{2.59}$$

then exactly the same indexed derivation shows that the total energy of the whole system is conserved.

Susskind closes this part of the lecture by coming back one last time to the harmonic oscillator. For the spring,

$$V(x) = \frac{k}{2} x^2, \tag{2.60}$$

so

$$E = \frac{1}{2} m \dot{x}^2 + \frac{k}{2} x^2. \tag{2.61}$$

When $m = k = 1$, this reduces to

$$E = \frac{1}{2} \left(\dot{x}^2 + x^2 \right). \tag{2.62}$$

But earlier we found

$$x^2 + p^2 = c^2, \tag{2.63}$$

and in the normalization $m = 1$ we have $p = \dot{x}$. Therefore the circles in phase space are nothing but contours of constant energy. The oscillator returns here not as a fresh example but as the payoff: the cycles in phase space are the level curves of the conserved quantity.

2.8 Summary

The lecture begins with Aristotle not to linger over a historical mistake, but to test what a law of motion should do. Aristotle's law becomes an update rule, predicts the future, and then reveals its weakness: distinct initial states collapse onto the same final rest state, so the past is not stably recoverable. Newton's law changes that by changing the very notion of state. Once we pass from position alone to phase space (x, p), the second-order equation becomes a pair of first-order equations, and prediction is restored in the proper sense.

From there the spring becomes the central example twice over. Under Aristotle it decays. Under Newton it oscillates. In phase space it runs on circles, and those circles announce conserved quantities. Newton's third law yields conservation of total momentum. Conservative forces yield conservation of energy. In one dimension the potential description is almost automatic; in several dimensions it is a substantive law of nature. That is where the lecture leaves us: with phase space, with conservation laws, and with the harmonic oscillator already hinting at the more general formulations of mechanics still to come.